JARED THRONE

(248) 345-1882 | jaredjthrone@gmail.com | linkedin.com/in/jared-throne-507138207

EDUCATION

Michigan State University, College of Engineering **May 2025**
Bachelor of Science, Mechanical Engineering | *GPA: 3.5 / 4.0*

PROFESSIONAL EXPERIENCE

Fluids Systems Engineer **February 2026 - Present**
Quest Defense | *Blue Ash, OH*

- Own design of fuel and hydraulic components (pumps, actuators, mechanical controls) for Fortune 500 customers; capture and incorporate system requirements into component designs.
- Manage suppliers across cost, performance, and manufacturability; design component interfaces and perform tolerance stack-up analysis.
- Present in technical and program reviews, interfacing directly with internal and external engineering groups to resolve complex design problems.

Research Intern **May - August 2024**
RWTH Aachen University | *Aachen, Germany*

- Executed time-resolved PIV experiments across four vessel geometries using blood analogous fluid and modeled geometry in CATIA.
- Applied the Visvalingam-Whyatt algorithm in MATLAB to smooth piston motion curves for pulsatile flow generation.
- Simulated internal flow using computational methods to validate experimental thermofluid findings across vessel geometry variations.

Process and Equipment Technician **November 2023 - May 2024**
Great Lakes Crystal Technologies | *East Lansing, MI*

- CNC machined molybdenum components to tight dimensional tolerances for CVD diamond growth reactor hardware, directly supporting production of single-crystal diamond substrates used in quantum sensing, AI chip packaging, and high-power electronics.
- Coordinated cross-functional troubleshooting to resolve equipment and process issues, minimizing downtime in an active diamond substrate manufacturing facility.

Systems Engineering Intern **May - October 2023**
Fraunhofer USA CMW | *East Lansing, MI*

- Designed water conditioning systems achieving 25 GPM flow rate for diamond growth thermal regulation within Fraunhofer's Coatings and Diamond Technologies Division, supporting CVD-based synthesis of diamond materials for quantum sensing and power electronics applications.
- Interpreted GD&T drawings for complex assemblies and developed safety systems improving workplace safety for 20+ employees.

PROJECTS

NASA Psyche Mission Rover **Dec 2024**
Designed rover mechanics in SolidWorks for asteroid 16-Psyche terrain; performed FEA, thermal, and structural simulation for component validation and rover dynamics.

Mini Segway, ME 456 **Apr 2025**
Tuned a PID controller in MATLAB Simulink to navigate a self-balancing segway through a competitive obstacle course design challenge.

MAC Launcher, ME 470 **Apr 2024**
Engineered a cornhole launcher using iterative design, FEA, and manufacturability constraints to optimize accuracy and ease of fabrication.

Vortex Shedding Visualization, ME 332 **Apr 2024**
Ran controlled flow-structure experiments with hydrogen bubble visualization; analyzed Reynolds and Strouhal number relationships.

TECHNICAL SKILLS

CAD: Siemens NX | SolidWorks | CATIA | AutoCAD

Simulation: FEA | Thermal Analysis | ANSYS Fluent | MATLAB | Simulink

Methods: GD&T | Tolerance Stack-Up Analysis | Design for Manufacturability | Experimental Design

Manufacturing: CNC Machining | Milling | Additive Manufacturing | 3D Printing | Mechanical Assembly

Programming: MATLAB | Python